CollegeBoard

AP® Calculus AB Practice Exam

From the 2016 Administration

CALCULUS AB
SECTION I, Part A
Time—55 minutes
Number of questions—28

A CALCULATOR MAY NOT BE USED ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

In this exam:

(1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

(2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix "arc" (e.g., $\sin^{-1} x = \arcsin x$).

GO ON TO THE NEXT PAGE.

1. If $y = \cos 2x$, then $\frac{dy}{dx} =$

(A) $-2\sin 2x$ (B) $-\sin 2x$ (C) $\sin 2x$ (D) $2\sin 2x$ (E) $2\sin x$

---

2. $\int x^2\left(x^3 - 1\right)^{10} dx =$

(A) $\frac{x^3}{3}\left(\frac{x^4}{4} - x\right)^{10} + C$

(B) $\frac{\left(x^3 - 1\right)^{11}}{11} + C$

(C) $\frac{x^2\left(x^3 - 1\right)^{11}}{11} + C$

(D) $\frac{\left(x^3 - 1\right)^{11}}{33} + C$

(E) $\frac{x^3\left(x^3 - 1\right)^{11}}{33} + C$

GO ON TO THE NEXT PAGE.

3. $\lim_{x \to \infty} \frac{\sqrt{9x^4 + 1}}{4x^2 + 3}$ is

(A) $\frac{1}{3}$ (B) $\frac{3}{4}$ (C) $\frac{3}{2}$ (D) $\frac{9}{4}$ (E) infinite

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4. If $y = \left(\frac{x}{x+1}\right)^5$, then $\frac{dy}{dx} =$

(A) $5(1+x)^4$ (B) $\frac{x^4}{(x+1)^4}$ (C) $\frac{5x^4}{(x+1)^4}$ (D) $\frac{5x^4}{(x+1)^6}$ (E) $\frac{5x^4(2x+1)}{(x+1)^6}$

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t (minutes)	0	4	7	9
$r(t)$ (gallons per minute)	9	6	4	3

5. Water is flowing into a tank at the rate $r(t)$, where $r(t)$ is measured in gallons per minute and t is measured in minutes. The tank contains 15 gallons of water at time $t = 0$. Values of $r(t)$ for selected values of t are given in the table above. Using a trapezoidal sum with the three intervals indicated by the table, what is the approximation of the number of gallons of water in the tank at time $t = 9$?

(A) 52 (B) 57 (C) 67 (D) 77 (E) 79

6. The slope of the line tangent to the graph of $y = \ln(1 - x)$ at $x = -1$ is

(A) -1 (B) $-\frac{1}{2}$ (C) $\frac{1}{2}$ (D) $\ln 2$ (E) 1

7. For which of the following pairs of functions f and g is $\lim_{x \to \infty} \frac{f(x)}{g(x)}$ infinite?

(A) $f(x) = x^2 + 2x$ and $g(x) = x^2 + \ln x$

(B) $f(x) = 3x^3$ and $g(x) = x^4$

(C) $f(x) = 3^x$ and $g(x) = x^3$

(D) $f(x) = 3e^x + x^3$ and $g(x) = 2e^x + x^2$

(E) $f(x) = \ln(3x)$ and $g(x) = \ln(2x)$

8. $\displaystyle\int_0^4 \frac{x}{\sqrt{x^2+9}}\,dx =$

(A) -2 (B) $-\frac{2}{15}$ (C) 1 (D) 2 (E) 5

---

9. Let f be the function with derivative given by $f'(x) = \dfrac{-2x}{\left(1+x^2\right)^2}$. On what interval is f decreasing?

(A) $[0, \infty)$ only

(B) $(-\infty, 0]$ only

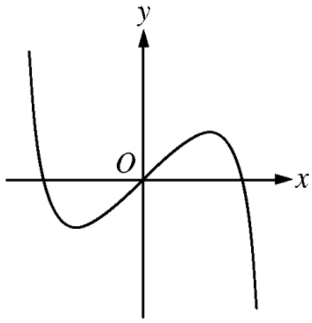
(C) $\left[-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$ only

(D) $(-\infty, \infty)$

(E) There is no such interval.

10. $\int \left(e^x + e\right) dx =$

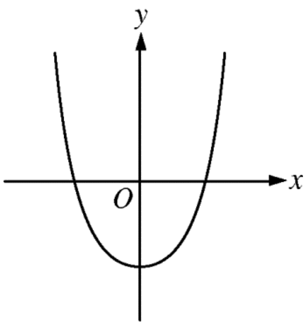
(A) $e^x + C$ (B) $2e^x + C$ (C) $e^x + e + C$ (D) $e^{x+1} + ex + C$ (E) $e^x + ex + C$



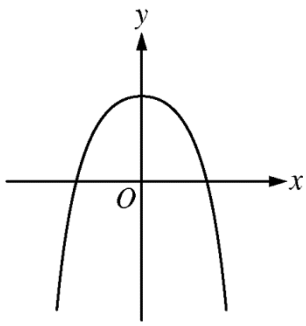
Graph of f

11. The graph of the function f is shown in the figure above. Which of the following could be the graph of f', the derivative of f?

(A)



(B)



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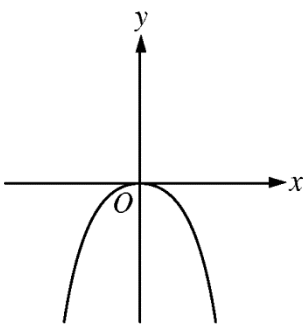

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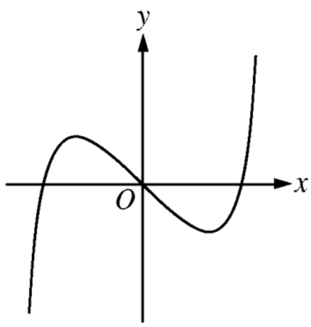

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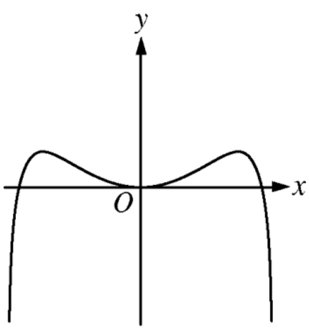

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12. If $0 < c < 1$, what is the area of the region enclosed by the graphs of $y = 0$, $y = \frac{1}{x}$, $x = c$, and $x = 1$?

(A) $\ln(1 - c)$ (B) $\ln\left(\frac{1}{c}\right)$ (C) $\ln c$ (D) $\frac{1}{c^2} - 1$ (E) $1 - \frac{1}{c^2}$

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13. $\frac{d}{dx}\left(\tan^{-1} x + 2\sqrt{x}\right) =$

(A) $-\frac{1}{\sin^2 x} + \frac{1}{\sqrt{x}}$

(B) $\frac{1}{\sqrt{1 - x^2}} - 4\sqrt[3]{x}$

(C) $\frac{1}{\sqrt{1 - x^2}} + \frac{1}{\sqrt{x}}$

(D) $\frac{1}{1 + x^2} - 4\sqrt[3]{x}$

(E) $\frac{1}{1 + x^2} + \frac{1}{\sqrt{x}}$

14. If $y = f(x)$ is a solution to the differential equation $\frac{dy}{dx} = e^{x^2}$ with the initial condition $f(0) = 2,$ which of the following is true?

(A) $f(x) = 1 + e^{x^2}$

(B) $f(x) = 2xe^{x^2}$

(C) $f(x) = \int_1^x e^{t^2}\,dt$

(D) $f(x) = 2 + \int_0^x e^{t^2}\,dt$

(E) $f(x) = 2 + \int_2^x e^{t^2}\,dt$

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15. A function $f(t)$ gives the rate of evaporation of water, in liters per hour, from a pond, where t is measured in hours since 12 noon. Which of the following gives the meaning of $\int_4^{10} f(t)\,dt$ in the context described?

(A) The total volume of water, in liters, that evaporated from the pond during the first 10 hours after 12 noon

(B) The total volume of water, in liters, that evaporated from the pond between 4 P.M. and 10 P.M.

(C) The net change in the rate of evaporation, in liters per hour, from the pond between 4 P.M. and 10 P.M.

(D) The average rate of evaporation, in liters per hour, from the pond between 4 P.M. and 10 P.M.

(E) The average rate of change in the rate of evaporation, in liters per hour per hour, from the pond between 4 P.M. and 10 P.M.

16. The first derivative of the function f is given by $f'(x) = 3x^4 - 12x^3$. What are the x-coordinates of the points of inflection of the graph of f ?

(A) $x = 3$ only

(B) $x = 4$ only

(C) $x = 0$ and $x = 2$

(D) $x = 0$ and $x = 3$

(E) $x = 0$ and $x = 4$

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17. Let f be the function defined by $f(x) = \frac{1}{x}$. What is the average value of f on the interval $[4, 6]$?

(A) $-\frac{1}{24}$ (B) $\frac{5}{24}$ (C) $\frac{1}{2}\ln\frac{3}{2}$ (D) $\ln\frac{3}{2}$ (E) $\frac{1}{2}\ln 2$

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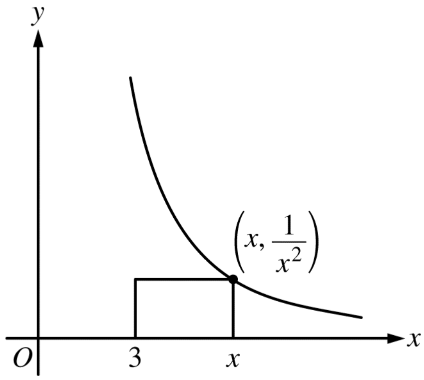

18. The points $(3, 0)$, $(x, 0)$, $\left(x, \frac{1}{x^2}\right)$, and $\left(3, \frac{1}{x^2}\right)$ are the vertices of a rectangle, where $x \geq 3$, as shown in the figure above. For what value of x does the rectangle have a maximum area?

(A) 3

(B) 4

(C) 6

(D) 9

(E) There is no such value of x.

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19. What are all values of x for which $\int_x^2 t^3\,dt$ is equal to 0 ?

(A) -2 only (B) 0 only (C) 2 only (D) -2 and 2 only (E) -2, 0, and 2

GO ON TO THE NEXT PAGE.

20. Let h be the function defined by $h(x) = \int_{\pi/4}^{x} \sin^2 t \, dt$. Which of the following is an equation for the line tangent to the graph of h at the point where $x = \frac{\pi}{4}$?

(A) $y = \frac{1}{2}$

(B) $y = \sqrt{2}\,x$

(C) $y = x - \frac{\pi}{4}$

(D) $y = \frac{1}{2}\left(x - \frac{\pi}{4}\right)$

(E) $y = \frac{\sqrt{2}}{2}\left(x - \frac{\pi}{4}\right)$

x	$f(x)$
–1	–30
0	–2
3	10
5	18

21. The table above gives selected values for a twice-differentiable function f. Which of the following must be true?

(A) f has no critical points in the interval $-1 < x < 5$.

(B) $f'(x) = 8$ for some value of x in the interval $-1 < x < 5$.

(C) $f'(x) > 0$ for all values of x in the interval $-1 < x < 5$.

(D) $f''(x) < 0$ for all values of x in the interval $-1 < x < 5$.

(E) The graph of f has no points of inflection in the interval $-1 < x < 5$.

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22. A particle moves along the x-axis so that at time $t \geq 0$, the acceleration of the particle is $a(t) = 15\sqrt{t}$. The position of the particle is 10 when $t = 0$, and the position of the particle is 20 when $t = 1$. What is the velocity of the particle at time $t = 0$?

(A) -14 (B) 0 (C) 5 (D) 6 (E) 10

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23. Which of the following is the solution to the differential equation $\frac{dy}{dx} = \frac{2xy}{x^2 + 1}$ whose graph contains the point $(0, 1)$?

(A) $y = e^{x^2}$

(B) $y = x^2 + 1$

(C) $y = \ln\left(x^2 + 1\right)$

(D) $y = 1 + \ln\left(x^2 + 1\right)$

(E) $y = \sqrt{1 + 2\ln\left(x^2 + 1\right)}$

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24. Sand is deposited into a pile with a circular base. The volume V of the pile is given by $V = \frac{r^3}{3}$, where r is the radius of the base, in feet. The circumference of the base is increasing at a constant rate of 5π feet per hour. When the circumference of the base is 8π feet, what is the rate of change of the volume of the pile, in cubic feet per hour?

(A) $\frac{8}{\pi}$ (B) 16 (C) 40 (D) 40π (E) 80π

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25. $\lim_{h \to 0} \frac{e^{-1-h} - e^{-1}}{h}$ is

(A) -1 (B) $\frac{-1}{e}$ (C) 0 (D) $\frac{1}{e}$ (E) nonexistent

26. Let f be the function given by $f(x) = x^3 + 5x$. For what value of x in the closed interval $[1,3]$ does the instantaneous rate of change of f equal the average rate of change of f on that interval?

(A) $\sqrt{\frac{7}{3}}$ (B) $\sqrt{\frac{13}{3}}$ (C) $\sqrt{5}$ (D) $\sqrt{6}$ (E) $\sqrt{\frac{19}{3}}$

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27. If $e^{xy} - y^2 = e - 4$, then at $x = \frac{1}{2}$ and $y = 2$, $\frac{dy}{dx} =$

(A) $\frac{e}{4}$ (B) $\frac{e}{2}$ (C) $\frac{4e}{8 - e}$ (D) $\frac{4e}{4 - e}$ (E) $\frac{8 - 4e}{e}$

28. Let f be the function defined by $f(x) = x^3 + x^2 + x$. Let $g(x) = f^{-1}(x)$, where $g(3) = 1$. What is the value of $g'(3)$?

(A) $\frac{1}{39}$ (B) $\frac{1}{34}$ (C) $\frac{1}{6}$ (D) $\frac{1}{3}$ (E) 39

END OF PART A OF SECTION I

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON PART A ONLY.

DO NOT GO ON TO PART B UNTIL YOU ARE TOLD TO DO SO.

CALCULUS AB

SECTION I, Part B

Time—50 minutes

Number of questions—17

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–92.

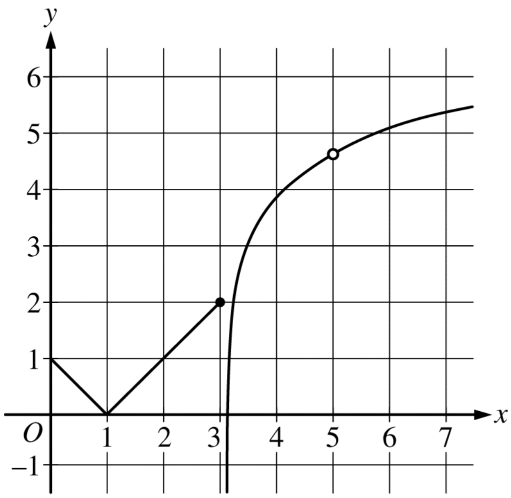
YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

(1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.

(2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

(3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix "arc" (e.g., $\sin^{-1} x = \arcsin x$).



Graph of f

76. The graph of a function f is shown above. Which of the following limits does not exist?

(A) $\lim\limits_{x \to 1^-} f(x)$ (B) $\lim\limits_{x \to 1} f(x)$ (C) $\lim\limits_{x \to 3^-} f(x)$ (D) $\lim\limits_{x \to 3} f(x)$ (E) $\lim\limits_{x \to 5} f(x)$

77. Let f be a function that is continuous on the closed interval $[1, 3]$ with $f(1) = 10$ and $f(3) = 18$. Which of the following statements must be true?

(A) $10 \le f(2) \le 18$

(B) f is increasing on the interval $[1, 3]$.

(C) $f(x) = 17$ has at least one solution in the interval $[1, 3]$.

(D) $f'(x) = 8$ has at least one solution in the interval $(1, 3)$.

(E) $\int_1^3 f(x)\,dx > 20$

78. Let R be the region bounded by the graphs of $y = e^x$, $y = e^3$, and $x = 0$. Which of the following gives the volume of the solid formed by revolving R about the line $y = -1$?

(A) $\pi \int_0^3 \left(e^3 - e^x + 1\right)^2 dx$

(B) $\pi \int_0^3 \left(e^3 - e^x - 1\right)^2 dx$

(C) $\pi \int_0^3 \left[\left(e^3 - e^x\right)^2 + 1\right] dx$

(D) $\pi \int_0^3 \left[\left(e^3 - e^x\right)^2 - 1\right] dx$

(E) $\pi \int_0^3 \left[\left(e^3 + 1\right)^2 - \left(e^x + 1\right)^2\right] dx$

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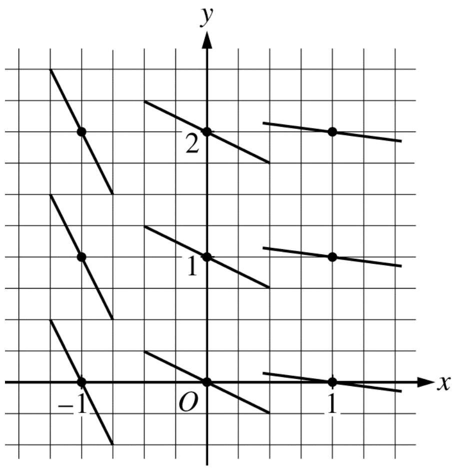
79. The number of people who have entered a museum on a certain day is modeled by a function $f(t)$, where t is measured in hours since the museum opened that day. The number of people who have left the museum since it opened that same day is modeled by a function $g(t)$. If $f'(t) = 380\left(1.02^t\right)$ and $g'(t) = 240 + 240\sin\left(\dfrac{\pi(t-4)}{12}\right)$, at what time t, for $1 \le t \le 11$, is the number of people in the museum at a maximum?

(A) 1 (B) 7.888 (C) 9.446 (D) 10.974 (E) 11

x	0	1	2	3
$f(x)$	5	2	3	6
$f'(x)$	-3	1	3	4

80. The derivative of the function f is continuous on the closed interval $[0, 4]$. Values of f and f' for selected values of x are given in the table above. If $\int_0^4 f'(t)\, dt = 8,$ then $f(4) =$

(A) 0 (B) 3 (C) 5 (D) 10 (E) 13



81. A slope field for a differential equation is shown in the figure above. If $y = f(x)$ is the particular solution to the differential equation through the point $(-1, 2)$ and $h(x) = 3x \cdot f(x)$, then $h'(-1) =$

(A) -6 (B) -2 (C) 0 (D) 1 (E) 12

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82. If f is a continuous function such that $f(2) = 6$, which of the following statements must be true?

(A) $\lim\limits_{x \to 1} f(2x) = 3$

(B) $\lim\limits_{x \to 2} f(2x) = 12$

(C) $\lim\limits_{x \to 2} \dfrac{f(x) - f(2)}{x - 2} = 6$

(D) $\lim\limits_{x \to 2} f\left(x^2\right) = 36$

(E) $\lim\limits_{x \to 2} \left(f(x)\right)^2 = 36$

83. A particle moves along a straight line with velocity given by $v(t) = 5 + e^{t/3}$ for time $t \geq 0$. What is the acceleration of the particle at time $t = 4$?

(A) 0.422 (B) 0.698 (C) 1.265 (D) 8.794 (E) 28.381

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84. A home uses fuel oil at the rate $r(t) = 10 + 8\sin\left(\frac{t}{60}\right)$ gallons per day, where t is the number of days from the beginning of the heating season. To the nearest gallon, what is the total amount of fuel oil used from $t = 0$ to $t = 60$ days?

(A) 7 gal (B) 14 gal (C) 600 gal (D) 821 gal (E) 1004 gal

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85. The function f is defined on the open interval $0.4 < x < 2.4$ and has first derivative f' given by $f'(x) = \sin\left(x^2\right)$. Which of the following statements are true?

I. f has a relative maximum on the interval $0.4 < x < 2.4$.

II. f has a relative minimum on the interval $0.4 < x < 2.4$.

III. The graph of f has two points of inflection on the interval $0.4 < x < 2.4$.

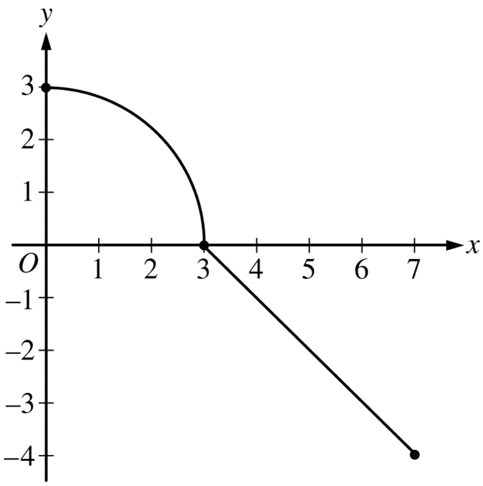
(A) I only

(B) II only

(C) III only

(D) I and III only

(E) II and III only



Graph of f

86. The graph of the function f, which has a domain of $[0, 7]$, is shown in the figure above. The graph consists of a quarter circle of radius 3 and a segment with slope -1. Let b be a positive number such that $\int_0^b f(x)\,dx = 0$. What is the value of b ?

(A) 3.760

(B) 5.548

(C) 5.659

(D) 6.760

(E) There is no such value of b.

87. The first derivative of the function g is given by $g'(x) = \cos\left(\pi x^2\right)$ for $-0.5 < x < 1.5$. On which of the following intervals is g decreasing?

(A) $-0.5 < x < 0$

(B) $0 < x < 1$

(C) $0.707 < x < 1.225$

(D) $1.225 < x < 1.414$

(E) $1.414 < x < 1.5$

88. The height above the ground of a passenger on a Ferris wheel t minutes after the ride begins is modeled by the differentiable function H, where $H(t)$ is measured in meters. Which of the following is an interpretation of the statement $H'(7.5) = 15.708$?

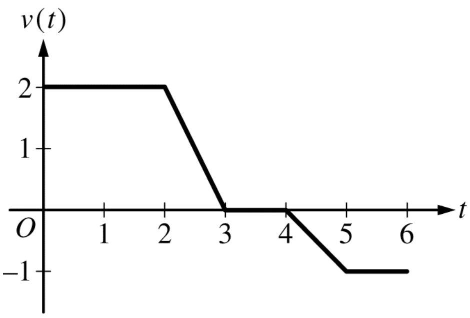
(A) The Ferris wheel is turning at a rate of 15.708 meters per minute when the passenger is 7.5 meters above the ground.

(B) The Ferris wheel is turning at a rate of 15.708 meters per minute 7.5 minutes after the ride begins.

(C) The passenger's height above the ground is increasing by 15.708 meters per minute when the passenger is 7.5 meters above the ground.

(D) The passenger's height above the ground is increasing by 15.708 meters per minute 7.5 minutes after the ride begins.

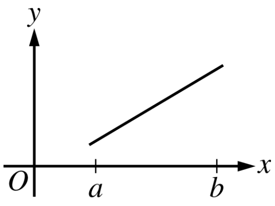
(E) The passenger is 15.708 meters above the ground 7.5 minutes after the ride begins.



89. A particle moves along a straight line for 6 seconds so that its velocity, in centimeters per second, is modeled by the graph shown. During the time interval $0 \le t \le 6$, what is the total distance the particle travels?

(A) 2 cm (B) 3.5 cm (C) 4 cm (D) 6.5 cm (E) 8.5 cm

90. Let f be a twice-differentiable function on the open interval (a, b). If $f'(x) > 0$ on (a, b) and $f''(x) < 0$ on (a, b), which of the following could be the graph of f?

(A)


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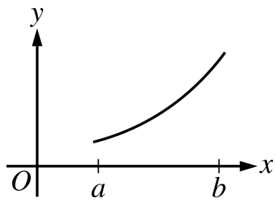
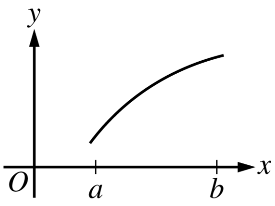


(C)


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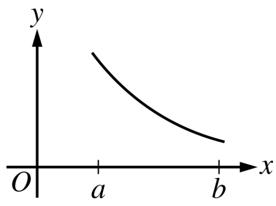

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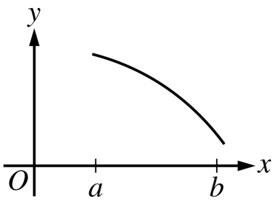

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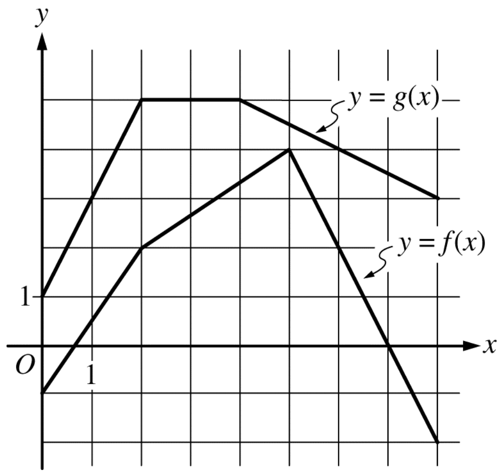

91. The graphs of f and g are shown above. If $h(x) = f(x)g(x)$, then $h'(6) =$

(A) -9 (B) -7 (C) 1 (D) 7 (E) 9

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92. In the xy-plane, the graph of the twice-differentiable function $y = f(x)$ is concave up on the open interval $(0, 2)$ and is tangent to the line $y = 3x - 2$ at $x = 1$. Which of the following statements must be true about the derivative of f?

(A) $f'(x) \le 3$ on the interval $(0.9, 1)$.

(B) $f'(x) \ge 3$ on the interval $(0.9, 1)$.

(C) $f'(x) < 0$ on the interval $(0.9, 1.1)$.

(D) $f'(x) > 0$ on the interval $(0.9, 1.1)$.

(E) $f'(x)$ is constant on the interval $(0.9, 1.1)$.

CALCULUS AB
SECTION II, Part A
Time—30 minutes
Number of problems—2

A graphing calculator is required for these problems.

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1. A company produces and sells chili powder. The company's weekly profit on the sale of x kilograms of chili powder is modeled by the function P given by $P(x) = 48x + 1.4x^2 - 0.05x^{2.8} - 270$, where $P(x)$ is in dollars and $0 \le x \le 80$.

   (a) Find the rate, in dollars per kilogram, at which the company's weekly profit is changing when it sells 32 kilograms of chili powder. Is the company's weekly profit increasing or decreasing when it sells 32 kilograms of chili powder? Give a reason for your answer.

   (b) How many kilograms of chili powder must the company sell to maximize its weekly profit? Justify your answer.

Continue problem 1 on page 5.

(c) The company plans to have a one-day sale on chili powder. Management estimates that t hours after the company store opens, chili powder will sell at a rate modeled by the function S given by $S(t) = 2 + \cos\left(\frac{\pi}{10}t^2\right)$ kilograms per hour. Based on this model, estimate the amount of chili powder, in kilograms, that will be sold during the first 5 hours of the sale.

(d) Using the function S from part (c), find the value of $S'(3)$. Interpret the meaning of this value in the context of the problem.

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t (weeks)	0	3	6	10	12
$G(t)$ (games per week)	160	450	900	2100	2400

2. A store tracks the sales of one of its popular board games over a 12-week period. The rate at which games are being sold is modeled by the differentiable function G, where $G(t)$ is measured in games per week and t is measured in weeks for $0 \le t \le 12$. Values of $G(t)$ are given in the table above for selected values of t.

(a) Approximate the value of $G'(8)$ using the data in the table. Show the computations that lead to your answer.

(b) Approximate the value of $\int_0^{12} G(t)\,dt$ using a right Riemann sum with the four subintervals indicated by the table. Explain the meaning of $\int_0^{12} G(t)\,dt$ in the context of this problem.

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Continue problem 2 on page 7.

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(c) One salesperson believes that, starting with 2400 games per week at time $t = 12,$ the rate at which games will be sold will increase at a constant rate of 100 games per week per week. Based on this model, how many total games will be sold in the 8 weeks between time $t = 12$ and $t = 20$?

(d) Another salesperson believes the best model for the rate at which games will be sold in the 8 weeks between time $t = 12$ and $t = 20$ is $M(t) = 2400e^{-0.01(t-12)^2}$ games per week. Based on this model, how many total games, to the nearest whole number, will be sold during this period?

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CALCULUS AB

SECTION II, Part B

Time—60 minutes

Number of problems—4

No calculator is allowed for these problems.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

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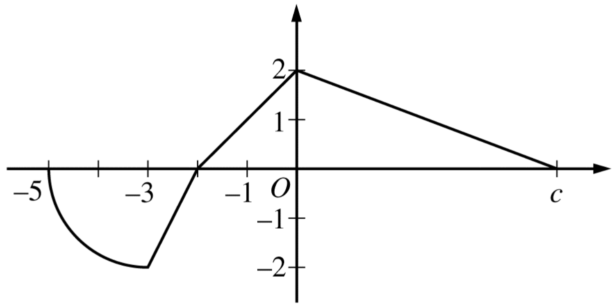

Graph of f

3. The function f is defined on the interval $-5 \le x \le c$, where $c > 0$ and $f(c) = 0$. The graph of f, which consists of three line segments and a quarter of a circle with center $(-3, 0)$ and radius 2, is shown in the figure above.

(a) Find the average rate of change of f over the interval $[-5, 0]$. Show the computations that lead to your answer.

(b) For $-5 \le x \le c$, let g be the function defined by $g(x) = \int_{-1}^{x} f(t)\,dt$. Find the x-coordinate of each point of inflection of the graph of g. Justify your answer.

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Continue problem 3 on page 15.

NO CALCULATOR ALLOWED

(c) Find the value of c for which the average value of f over the interval $-5 \le x \le c$ is $\frac{1}{2}$.

(d) Assume $c > 3$. The function h is defined by $h(x) = f\left(\frac{x}{2}\right)$. Find $h'(6)$ in terms of c.

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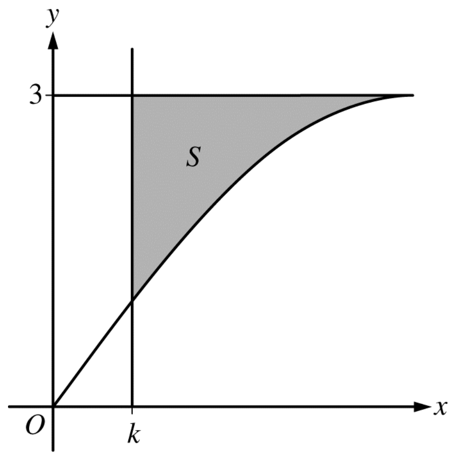

4. Let S be the shaded region in the first quadrant bounded above by the horizontal line $y = 3$, below by the graph of $y = 3\sin x$, and on the left by the vertical line $x = k$, where $0 < k < \frac{\pi}{2}$, as shown in the figure above.

(a) Find the area of S when $k = \frac{\pi}{3}$.

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Continue problem 4 on page 17.

NO CALCULATOR ALLOWED

(b) The area of S is a function of k. Find the rate of change of the area of S with respect to k when $k = \frac{\pi}{6}$.

(c) Region S is revolved about the horizontal line $y = 5$ to form a solid. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid when $k = \frac{\pi}{4}$.

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5. For $0 \le t \le 24$ hours, the temperature inside a refrigerator in a kitchen is given by the function W that satisfies the differential equation $\frac{dW}{dt} = \frac{3\cos t}{2W}$. $W(t)$ is measured in degrees Celsius $(°\text{C})$, and t is measured in hours. At time $t = 0$ hours, the temperature inside the refrigerator is $3°\text{C}$.

(a) Write an equation for the line tangent to the graph of $y = W(t)$ at the point where $t = 0$. Use the equation to approximate the temperature inside the refrigerator at $t = 0.4$ hour.

(b) Find $y = W(t)$, the particular solution to the differential equation with initial condition $W(0) = 3$.

Continue problem 5 on page 19.

NO CALCULATOR ALLOWED

(c) The temperature in the kitchen remains constant at $20°C$ for $0 \le t \le 24$. The cost of operating the refrigerator accumulates at the rate of \$0.001 per hour for each degree that the temperature in the kitchen exceeds the temperature inside the refrigerator. Write, but do not evaluate, an expression involving an integral that can be used to find the cost of operating the refrigerator for the 24-hour interval.

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NO CALCULATOR ALLOWED

$$f(x) = \begin{cases} 10 - 2x - x^2 & \text{for } x \le 1 \\ 3 + 4e^{x-1} & \text{for } x > 1 \end{cases}$$

6. Let f be the function defined above.

   (a) Is f continuous at $x = 1$? Why or why not?

   (b) Find the absolute minimum value and the absolute maximum value of f on the closed interval $-2 \le x \le 2$. Show the analysis that leads to your conclusion.

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Continue problem 6 on page 21.

NO CALCULATOR ALLOWED

(c) Find the value of $\int_0^2 f(x)\,dx$.

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Answer Key for AP Calculus AB Practice Exam, Section I

Question 1: A

Question 2: D

Question 3: B

Question 4: D

Question 5: C

Question 6: B

Question 7: C

Question 8: D

Question 9: A

Question 10: E

Question 11: B

Question 12: B

Question 13: E

Question 14: D

Question 15: B

Question 16: A

Question 17: C

Question 18: C

Question 19: D

Question 20: D

Question 21: B

Question 22: D

Question 23: B

Question 24: C

Question 25: B

Question 26: B

Question 27: C

Question 28: C

Question 76: D

Question 77: C

Question 78: E

Question 79: B

Question 80: E

Question 81: E

Question 82: E

Question 83: C

Question 84: D

Question 85: D

Question 86: D

Question 87: C

Question 88: D

Question 89: D

Question 90: C

Question 91: A

Question 92: A

2016 AP Calculus AB

Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Topic	Key	% Correct
1	Derivatives	A	86
2	Integrals	D	55
3	Functions, Graphs, And Limits	B	60
4	Derivatives	D	61
5	Integrals	C	42
6	Derivatives	B	54
7	Functions, Graphs, And Limits	C	34
8	Integrals	D	52
9	Derivatives	A	70
10	Integrals	E	57
11	Derivatives	B	72
12	Integrals	B	32
13	Derivatives	E	61
14	Integrals	D	60
15	Integrals	B	72
16	Derivatives	A	37
17	Integrals	C	44
18	Derivatives	C	47
19	Integrals	D	56
20	Integrals	D	49
21	Derivatives	B	43
22	Integrals	D	17
23	Integrals	B	33
24	Derivatives	C	29
25	Derivatives	B	25
26	Derivatives	B	46
27	Derivatives	C	39
28	Derivatives	C	37
76	Functions, Graphs, And Limits	D	70
77	Functions, Graphs, And Limits	C	63
78	Integrals	E	71
79	Derivatives	B	45
80	Integrals	E	45
81	Derivatives	E	29
82	Functions, Graphs, And Limits	E	49
83	Derivatives	C	82
84	Integrals	D	86
85	Derivatives	D	55

2016 AP Calculus AB

Question Descriptors and Performance Data

Question	Topic	Key	% Correct
86	Integrals	D	41
87	Derivatives	C	80
88	Derivatives	D	59
89	Integrals	D	57
90	Derivatives	C	76
91	Derivatives	A	51
92	Derivatives	A	27

AP® CALCULUS AB
2016 SCORING GUIDELINES

Question 1

A company produces and sells chili powder. The company's weekly profit on the sale of x kilograms of chili powder is modeled by the function P given by $P(x) = 48x + 1.4x^2 - 0.05x^{2.8} - 270$, where $P(x)$ is in dollars and $0 \le x \le 80$.

(a) Find the rate, in dollars per kilogram, at which the company's weekly profit is changing when it sells 32 kilograms of chili powder. Is the company's weekly profit increasing or decreasing when it sells 32 kilograms of chili powder? Give a reason for your answer.

(b) How many kilograms of chili powder must the company sell to maximize its weekly profit? Justify your answer.

(c) The company plans to have a one-day sale on chili powder. Management estimates that t hours after the company store opens, chili powder will sell at a rate modeled by the function S given by $S(t) = 2 + \cos\left(\frac{\pi}{10}t^2\right)$ kilograms per hour. Based on this model, estimate the amount of chili powder, in kilograms, that will be sold during the first 5 hours of the sale.

(d) Using the function S from part (c), find the value of $S'(3)$. Interpret the meaning of this value in the context of the problem.

(a) $P'(32) = 65.920$

Since $P'(32) > 0$, the company's profit is increasing when it sells 32 kilograms of chili powder.

$2 : \begin{cases} 1 : P'(32) \\ 1 : \text{increasing with reason} \end{cases}$

(b) $P'(x) = 48 + 2.8x - 0.14x^{1.8} = 0$ when $x = 58.358152$

x	$P(x)$
0	-270
58.358152	2893.04
80	1873.32

The company must sell 58.358 kilograms of chili powder to maximize its profit.

$3 : \begin{cases} 1 : \text{sets } P'(x) = 0 \\ 1 : \text{answer} \\ 1 : \text{justification} \end{cases}$

(c) Based on this model, the estimate, in kilograms, is

$\int_0^5 S(t)\,dt = 11.433$ (or 11.432).

$2 : \begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(d) $S'(3) = -0.582$

The rate of sale of chili powder is decreasing at a rate of 0.582 kilogram per hour per hour at time $t = 3$ hours.

$2 : \begin{cases} 1 : S'(3) \\ 1 : \text{interpretation} \end{cases}$

Question 2

t (weeks)	0	3	6	10	12
$G(t)$ (games per week)	160	450	900	2100	2400

A store tracks the sales of one of its popular board games over a 12-week period. The rate at which games are being sold is modeled by the differentiable function G, where $G(t)$ is measured in games per week and t is measured in weeks for $0 \le t \le 12$. Values of $G(t)$ are given in the table above for selected values of t.

(a) Approximate the value of $G'(8)$ using the data in the table. Show the computations that lead to your answer.

(b) Approximate the value of $\int_0^{12} G(t)\,dt$ using a right Riemann sum with the four subintervals indicated by the table. Explain the meaning of $\int_0^{12} G(t)\,dt$ in the context of this problem.

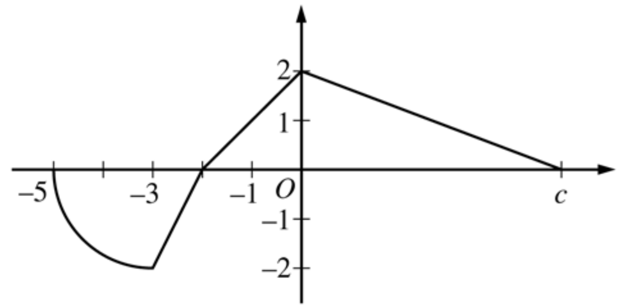
(c) One salesperson believes that, starting with 2400 games per week at time $t = 12$, the rate at which games will be sold will increase at a constant rate of 100 games per week per week. Based on this model, how many total games will be sold in the 8 weeks between time $t = 12$ and $t = 20$?

(d) Another salesperson believes the best model for the rate at which games will be sold in the 8 weeks between time $t = 12$ and $t = 20$ is $M(t) = 2400e^{-0.01(t-12)^2}$ games per week. Based on this model, how many total games, to the nearest whole number, will be sold during this period?

Solution	Scoring
(a) $G'(8) \approx \dfrac{G(10) - G(6)}{10 - 6} = \dfrac{2100 - 900}{10 - 6} = 300$ games per week per week	1 : approximation
(b) $\int_0^{12} G(t)\,dt \approx 3 \cdot G(3) + 3 \cdot G(6) + 4 \cdot G(10) + 2 \cdot G(12)$ $= 3 \cdot 450 + 3 \cdot 900 + 4 \cdot 2100 + 2 \cdot 2400$ $= 17250$ $\int_0^{12} G(t)\,dt$ represents the total number of games sold over the 12-week period $0 \le t \le 12$.	$3 : \begin{cases} 1 : \text{right Riemann sum} \\ 1 : \text{approximation} \\ 1 : \text{explanation} \end{cases}$
(c) The rate of sales of the game for $12 \le t \le 20$ is $R(t) = 2400 + 100(t - 12)$. Based on this model, the total number of games that will be sold between time $t = 12$ and $t = 20$ is $\int_{12}^{20} R(t)\,dt = 22400$.	$3 : \begin{cases} 1 : \text{rate function} \\ 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$
(d) $\int_{12}^{20} M(t)\,dt = 15784.07655$ Based on this model, the total number of games that will be sold between time $t = 12$ and $t = 20$ is 15784.	$2 : \begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

Question 3

The function f is defined on the interval $-5 \le x \le c$, where $c > 0$ and $f(c) = 0$. The graph of f, which consists of three line segments and a quarter of a circle with center $(-3, 0)$ and radius 2, is shown in the figure above.



Graph of f

(a) Find the average rate of change of f over the interval $[-5, 0]$. Show the computations that lead to your answer.

(b) For $-5 \le x \le c$, let g be the function defined by $g(x) = \int_{-1}^{x} f(t)\,dt$. Find the x-coordinate of each point of inflection of the graph of g. Justify your answer.

(c) Find the value of c for which the average value of f over the interval $-5 \le x \le c$ is $\frac{1}{2}$.

(d) Assume $c > 3$. The function h is defined by $h(x) = f\left(\frac{x}{2}\right)$. Find $h'(6)$ in terms of c.

(a) The average rate of change of f over the interval $[-5, 0]$ is

$$\frac{f(0) - f(-5)}{0 - (-5)} = \frac{2}{5}.$$

1 : answer

(b) $g'(x) = f(x)$

The graph of g has a point of inflection at $x = -3$ because $g' = f$ changes from decreasing to increasing at this point.

The graph of g has a point of inflection at $x = 0$ because $g' = f$ changes from increasing to decreasing at this point.

$$3 : \begin{cases} 1 : g'(x) = f(x) \\ 1 : \text{identifies } x = -3 \text{ and } x = 0 \\ 1 : \text{justification} \end{cases}$$

(c) $\frac{1}{c+5}\int_{-5}^{c} f(x)\,dx = \frac{1}{2}$

$\frac{1}{c+5}(-\pi + (-1) + 2 + c) = \frac{1}{2}$

$c = 3 + 2\pi$

$$3 : \begin{cases} 1 : \text{integral} \\ 1 : \text{equation} \\ 1 : \text{answer} \end{cases}$$

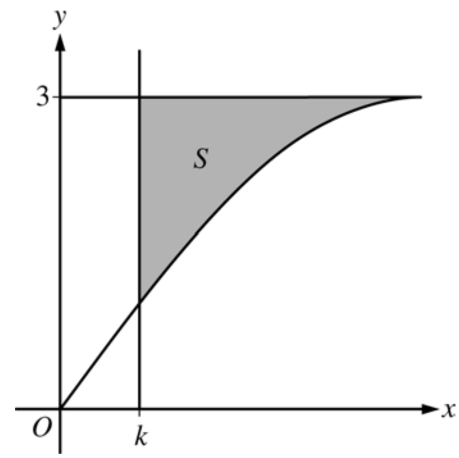
(d) $h'(x) = \frac{1}{2} f'\left(\frac{x}{2}\right)$

$h'(6) = \frac{1}{2} f'(3) = \frac{1}{2} \cdot \frac{-2}{c} = -\frac{1}{c}$

$$2 : \begin{cases} 1 : h'(x) \\ 1 : \text{answer} \end{cases}$$

Question 4

Let S be the shaded region in the first quadrant bounded above by the horizontal line $y = 3$, below by the graph of $y = 3\sin x$, and on the left by the vertical line $x = k$, where $0 < k < \frac{\pi}{2}$, as shown in the figure above.



(a) Find the area of S when $k = \frac{\pi}{3}$.

(b) The area of S is a function of k. Find the rate of change of the area of S with respect to k when $k = \frac{\pi}{6}$.

(c) Region S is revolved about the horizontal line $y = 5$ to form a solid. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid when $k = \frac{\pi}{4}$.

(a) $$\text{Area} = \int_{\pi/3}^{\pi/2} (3 - 3\sin x)\, dx = \Big[3x + 3\cos x\Big]_{\pi/3}^{\pi/2}$$
$$= 3\left[\left(\frac{\pi}{2} + 0\right) - \left(\frac{\pi}{3} + \frac{1}{2}\right)\right] = \frac{\pi}{2} - \frac{3}{2}$$

$3 : \begin{cases} 1 : \text{integrand} \\ 1 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$

(b) Let $A(k)$ be the area of S.
$$A(k) = \int_{k}^{\pi/2} (3 - 3\sin x)\, dx$$
$$A'(k) = -3 + 3\sin k$$
$$A'\left(\frac{\pi}{6}\right) = -3 + 3\sin\left(\frac{\pi}{6}\right) = -\frac{3}{2}$$

$3 : \begin{cases} 1 : \text{expression for area} \\ 1 : \text{expression for } A'(k) \\ 1 : \text{answer} \end{cases}$

(c) $$\text{Volume} = \pi\int_{\pi/4}^{\pi/2} \left[(5 - 3\sin x)^2 - (5 - 3)^2\right] dx$$
$$= \pi\int_{\pi/4}^{\pi/2} \left[(5 - 3\sin x)^2 - 4\right] dx$$

$3 : \begin{cases} 2 : \text{integrand} \\ 1 : \text{limits and constant} \end{cases}$

Question 5

For $0 \le t \le 24$ hours, the temperature inside a refrigerator in a kitchen is given by the function W that satisfies the differential equation $\frac{dW}{dt} = \frac{3\cos t}{2W}$. $W(t)$ is measured in degrees Celsius $(°\text{C})$, and t is measured in hours. At time $t = 0$ hours, the temperature inside the refrigerator is $3°\text{C}$.

(a) Write an equation for the line tangent to the graph of $y = W(t)$ at the point where $t = 0$. Use the equation to approximate the temperature inside the refrigerator at $t = 0.4$ hour.

(b) Find $y = W(t)$, the particular solution to the differential equation with initial condition $W(0) = 3$.

(c) The temperature in the kitchen remains constant at $20°\text{C}$ for $0 \le t \le 24$. The cost of operating the refrigerator accumulates at the rate of \$0.001 per hour for each degree that the temperature in the kitchen exceeds the temperature inside the refrigerator. Write, but do not evaluate, an expression involving an integral that can be used to find the cost of operating the refrigerator for the 24-hour interval.

Solution	Scoring
(a) $\left.\frac{dW}{dt}\right\rvert_{(t,\,W)=(0,\,3)} = \frac{3\cos 0}{2(3)} = \frac{1}{2}$ An equation for the tangent line is $y = \frac{1}{2}t + 3$. $W(0.4) \approx \frac{1}{2}(0.4) + 3 = 3.2°\text{C}$	$2 : \begin{cases} 1 : \text{tangent line equation} \\ 1 : \text{approximation} \end{cases}$
(b) $2W\ dW = 3\cos t\ dt$ $\int 2W\ dW = \int 3\cos t\ dt$ $W^2 = 3\sin t + C$ $3^2 = 3\sin 0 + C \Rightarrow C = 9$ $W^2 = 3\sin t + 9$ Since $W(0) = 3$, $W = \sqrt{3\sin t + 9}$ for $0 \le t \le 24$.	$5 : \begin{cases} 1 : \text{separation of variables} \\ 2 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ \quad\ \text{and uses initial condition} \\ 1 : \text{solves for } W \end{cases}$ Note: max $3/5$ [1-2-0-0] if no constant of integration Note: $0/5$ if no separation of variables
(c) $0.001\int_0^{24} (20 - W(t))\ dt$ dollars	$2 : \begin{cases} 1 : \text{integrand} \\ 1 : \text{limits and constant} \end{cases}$

Question 6

$$f(x) = \begin{cases} 10 - 2x - x^2 & \text{for } x \le 1 \\ 3 + 4e^{x-1} & \text{for } x > 1 \end{cases}$$

Let f be the function defined above.

(a) Is f continuous at $x = 1$? Why or why not?

(b) Find the absolute minimum value and the absolute maximum value of f on the closed interval $-2 \le x \le 2$. Show the analysis that leads to your conclusion.

(c) Find the value of $\int_0^2 f(x)\,dx$.

(a) $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-}\left(10 - 2x - x^2\right) = 7$

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+}\left(3 + 4e^{x-1}\right) = 7$

Therefore, $\lim_{x\to 1} f(x) = 7$.

Since $\lim_{x\to 1} f(x) = f(1)$, f is continuous at $x = 1$.

$2 : \begin{cases} 1 : \text{considers one-sided limits} \\ 1 : \text{answer with explanation} \end{cases}$

(b) For $x < 1$, $f'(x) = -2 - 2x$.

$f'(x) = 0 \Rightarrow x = -1$

For $x > 1$, $f'(x) = 4e^{x-1} \ne 0$.

At $x = 1$, $f'(x)$ is not defined.

x	$f(x)$
-2	10
-1	11
1	7
2	$3 + 4e$

The absolute minimum value of f on the interval $-2 \le x \le 2$ is 7. The absolute maximum value of f on the interval $-2 \le x \le 2$ is $3 + 4e$.

$4 : \begin{cases} 1 : \frac{d}{dx}\left(10 - 2x - x^2\right) \text{ and } \frac{d}{dx}\left(3 + 4e^{x-1}\right) \\ 1 : \text{identifies } x = -1 \text{ and } x = 1 \text{ as critical points} \\ 1 : \text{evaluates } f \text{ at endpoints} \\ 1 : \text{answers with analysis} \end{cases}$

(c)
$$\begin{aligned} \int_0^2 f(x)\,dx &= \int_0^1 \left(10 - 2x - x^2\right) dx + \int_1^2 \left(3 + 4e^{x-1}\right) dx \\ &= \left[10x - x^2 - \frac{1}{3}x^3\right]_0^1 + \left[3x + 4e^{x-1}\right]_1^2 \\ &= \left(10 - 1 - \frac{1}{3}\right) + \left[(6 + 4e) - (3 + 4)\right] \\ &= \frac{23}{3} + 4e \end{aligned}$$

$3 : \begin{cases} 1 : \text{sum of integrals} \\ 1 : \text{antiderivatives} \\ 1 : \text{answer} \end{cases}$